

Novel Death Association-Reaction Score Measure from Sentence Task to Predict Depression and Suicidal Ideation

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Introduction



- Accurate and timely assessment of depression and suicide risk is essential in developing and offering effective treatment.
- Current methods for assessing individual mental health and potential risk factors predominantly rely on self-reports or behavioral interviews.

Objective and hypothesis

- **Objective:** Investigate how reaction times (RTs) can reveal cognitive and behavioral patterns related to themes such as death, hopelessness, and suicidal ideation across control, depressed, and suicidal groups.
- **Hypothesis:** Individuals with suicidal ideation will demonstrate faster RTs for stimuli aligned with death-related or suicidal themes, reflecting stronger cognitive associations with these concepts compared to control and depressed groups.

Sentence Task- Experimental Design

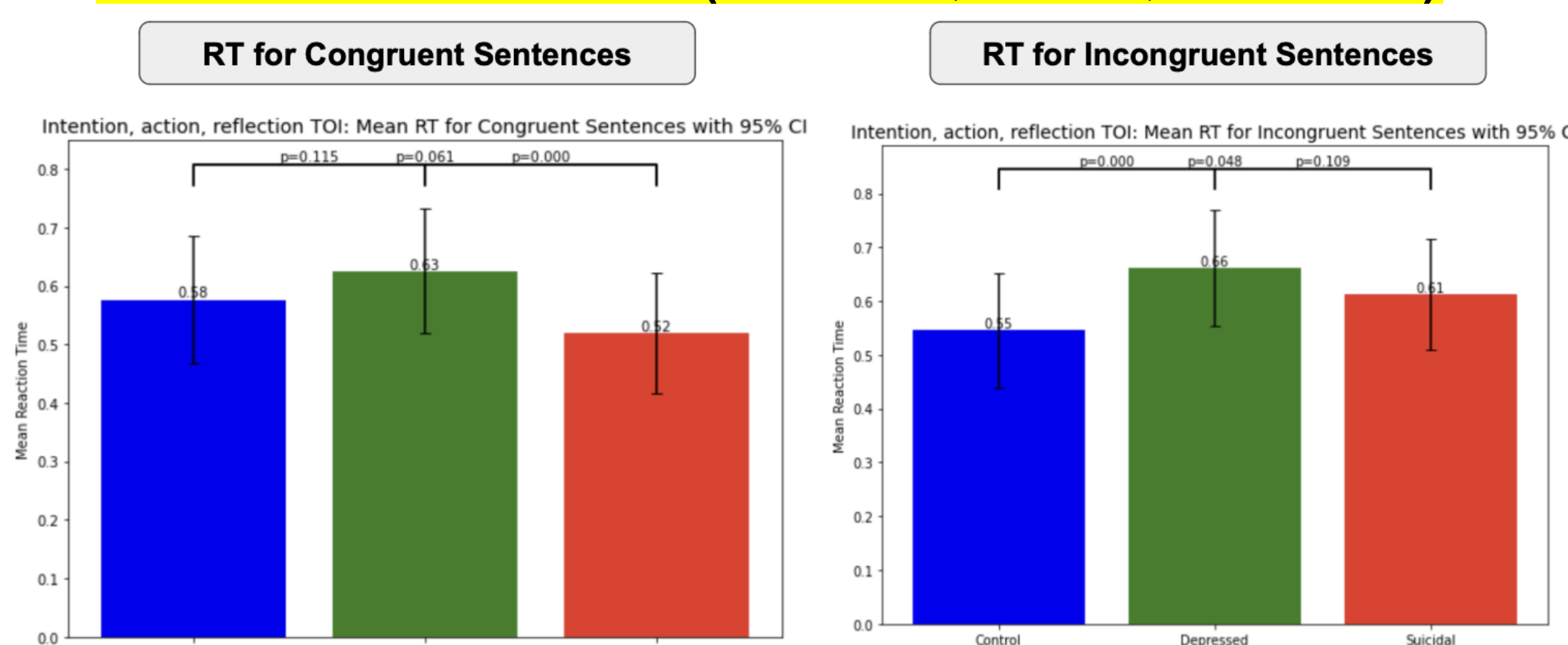
- Subjects completed a sentence task (*agree/disagree*) while their RTs were recorded
- Sentences drawn from *four Topics of Interest (TOIs)*: Biographical, Action, Reflection, and Intention.
- Each TOI included 20 pairs of statement sentences.
- Sentences were presented in a *first-person form* and predicted to be congruent (eliciting agreement) or incongruent (eliciting disagreement) for individuals within the three mental health groups (Control, Depressed and Suicidal).
- **Congruent:** The Subject's response is in **agreement** with the sentence.
- **Incongruent:** The Subject's response is in **disagreement** with the sentence.
- Total of **146 Subjects**: Control (n= 49), Depressed (n= 47), and Suicidal (n= 50) groups based on their PHQ-9 and SIS (Suicidal Ideation Scale) scores.

Novel DAR-measure for suicidal ideation

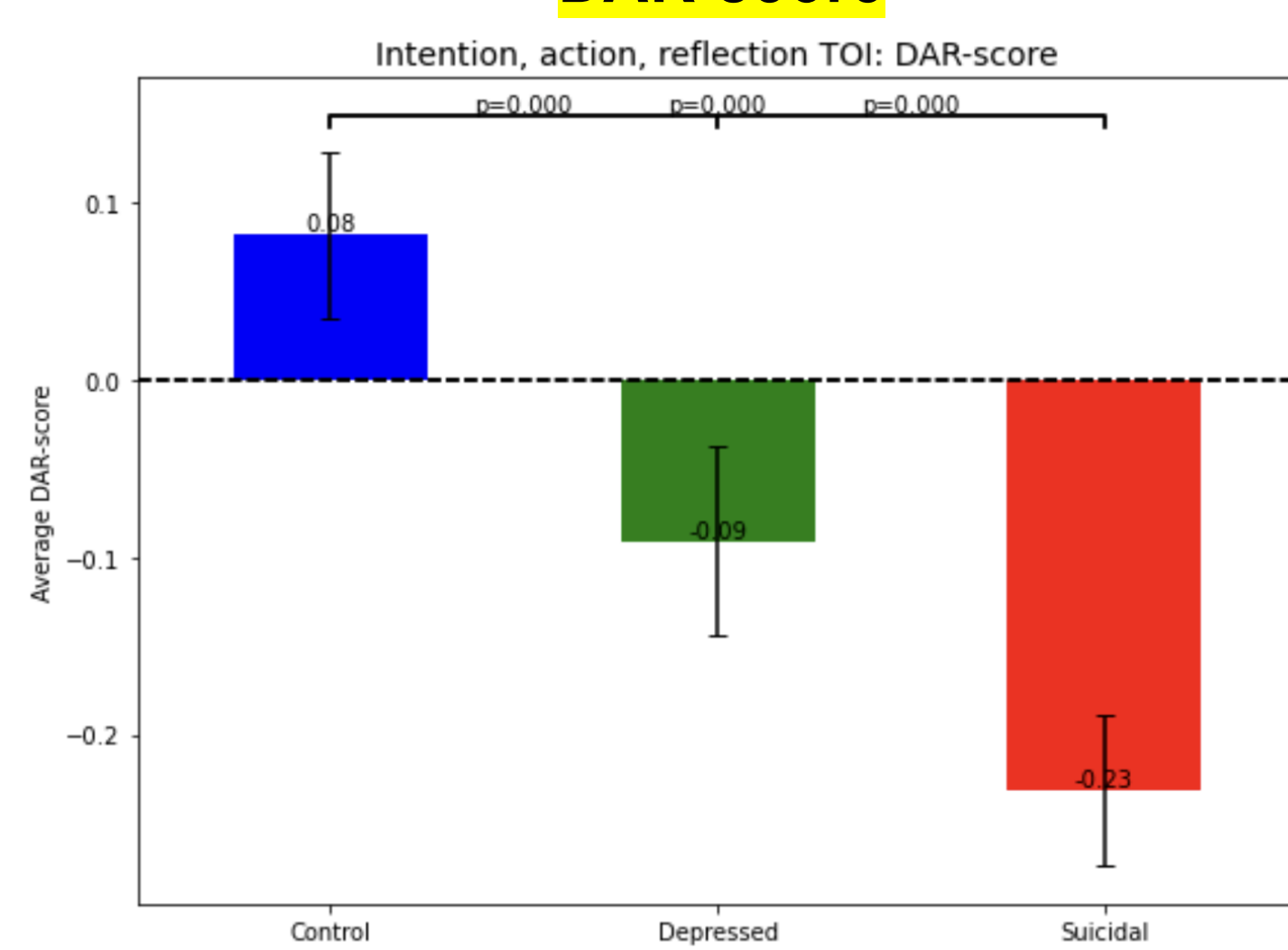
$$\text{DAR-score} = \frac{(\text{Mean RT} [\text{congruent sentences}] - \text{Mean RT} [\text{incongruent sentences}])}{\text{SD RT} [\text{all sentences}]}$$

- Premise: DAR-score is expected to reveal and capture associations with depression and suicidal stimuli

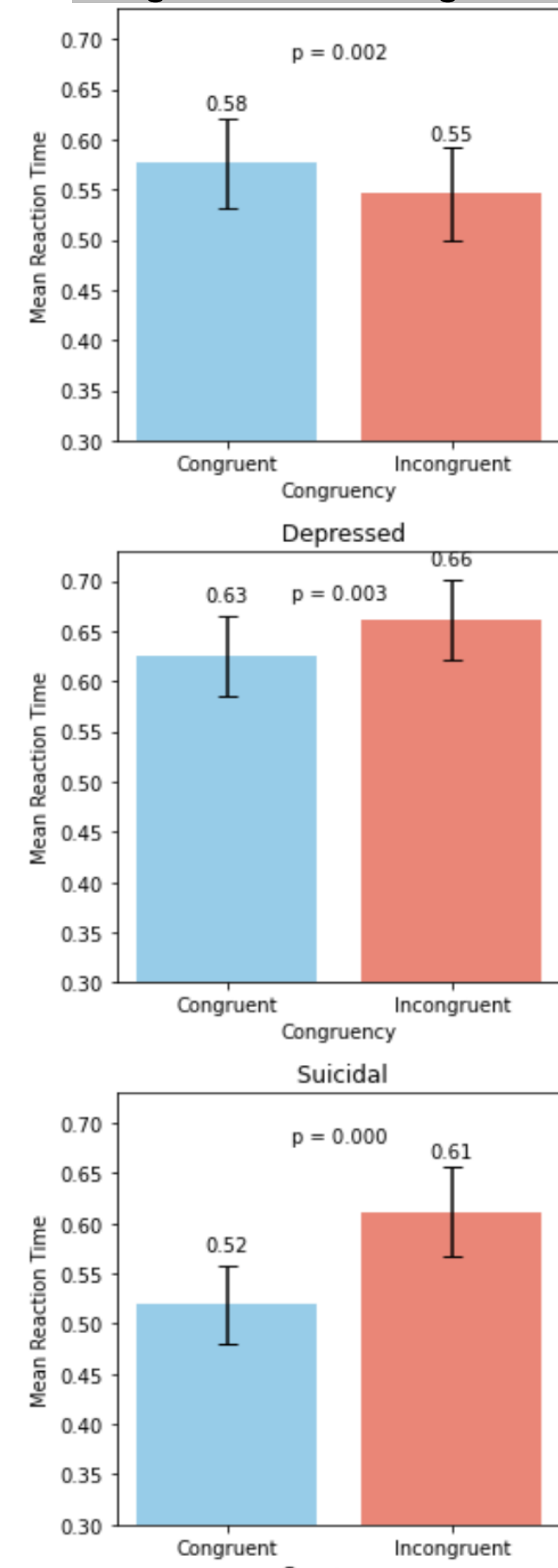
Combination of 3 TOIs (Intention, action, reflection)



DAR-score

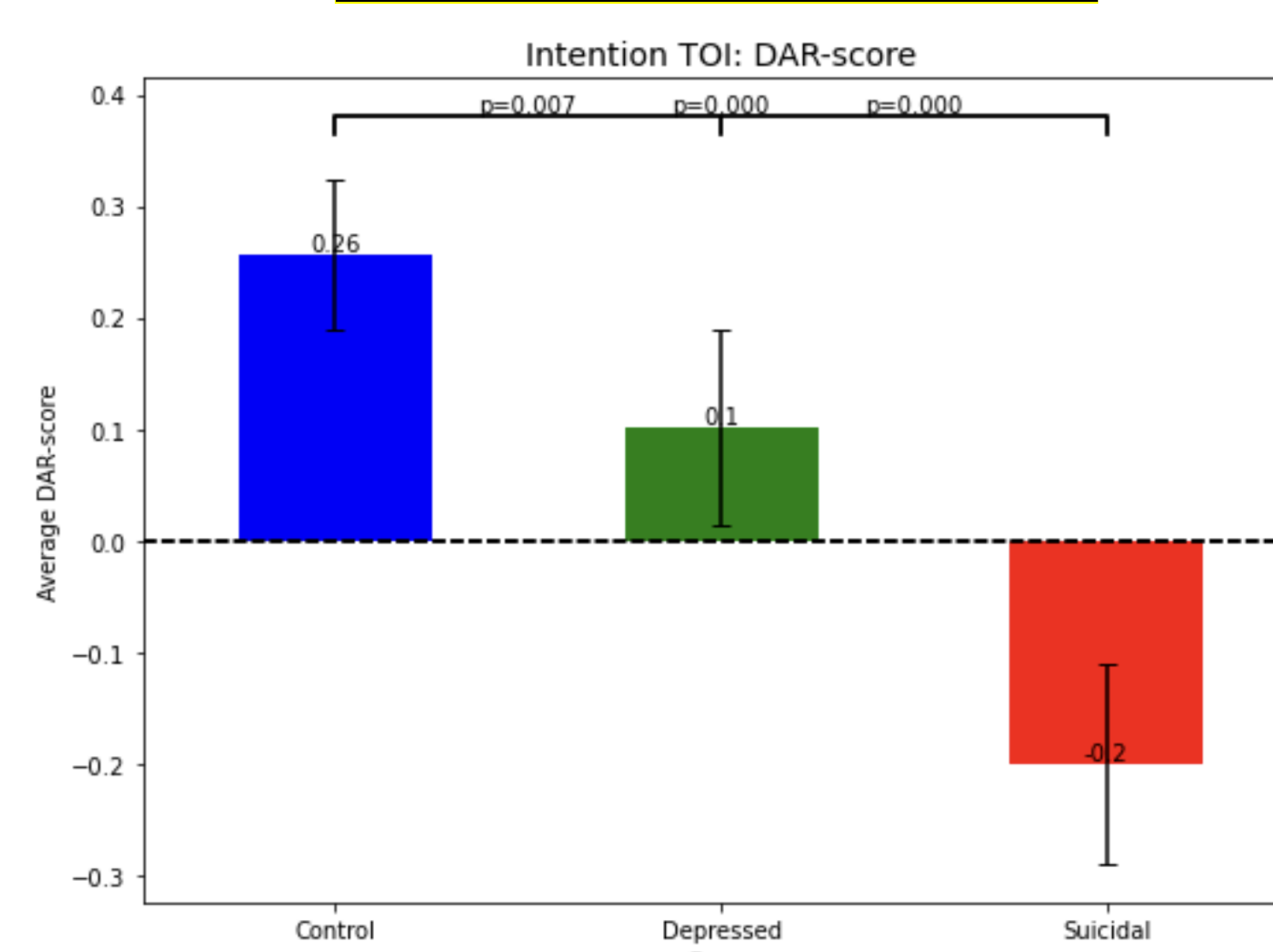


RTs: congruent vs Incongruent

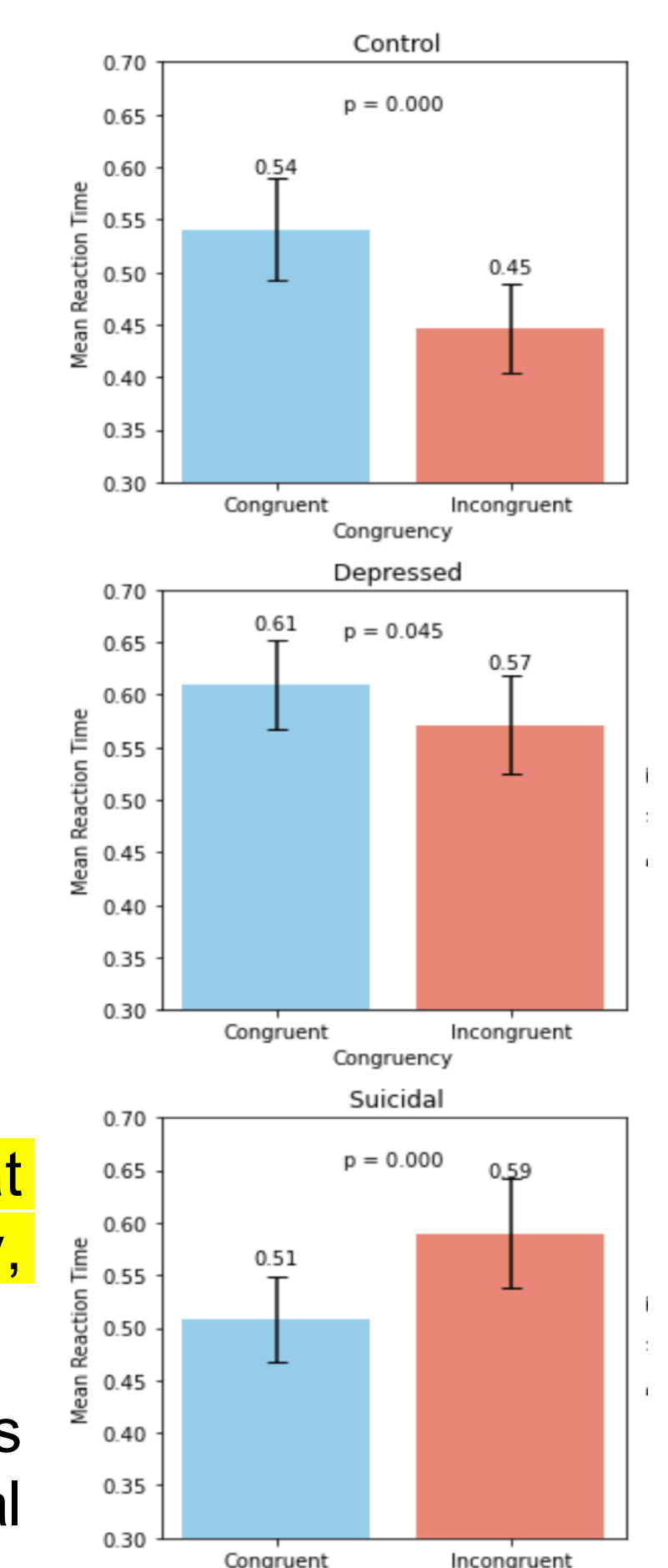


Analysis Results

DAR-score: Intention TOI



RTs: congruent vs Incongruent



- The suicidal group has smaller DAR values indicating that suicidal attributes are strongly associated with their memory, leading to faster RT for congruent stimuli
- DAR-score captures suicidal and death-related associations and effectively distinguishes between control and suicidal groups.

Prediction Results

- **Classifier:** Logistic Regression with DAR-score as a feature
- **Stratified 5-fold cross-validation:** In each fold, 117 subjects were used for training and 29 for testing, ensuring all subjects were tested.

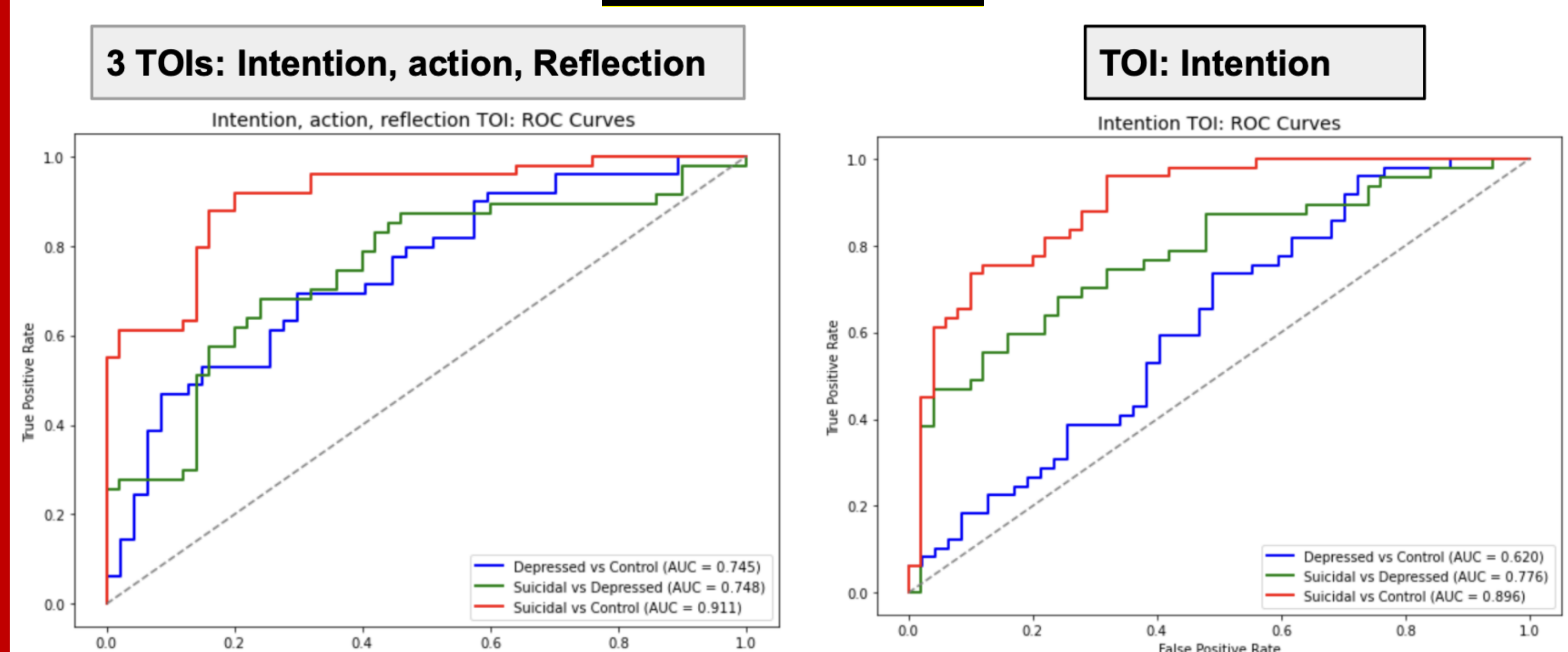
Intention TOI

Setting	Balanced Accuracy	Sensitivity	Specificity	F1-score	AUC
Control vs Depressed	61 ± 9	0.49 ± 0.16	0.72 ± 0.13	0.54 ± 0.13	0.62 ± 0.9
Depressed vs Suicidal	71 ± 11	0.72 ± 0.13	0.69 ± 0.13	0.72 ± 0.11	0.78 ± 0.12
Control vs Suicidal	78 ± 8	0.75 ± 0.13	0.81 ± 0.13	0.77 ± 0.08	0.90 ± 0.06

Combination of 3 TOIs (Intention, action, reflection)

Setting	Balanced Accuracy	Sensitivity	Specificity	F1-score	AUC
Control vs Depressed	64 ± 7	0.55 ± 0.15	0.74 ± 0.12	0.59 ± 0.10	0.75 ± 0.08
Depressed vs Suicidal	69 ± 8	0.81 ± 0.11	0.58 ± 0.15	0.73 ± 0.07	0.74 ± 0.10
Control vs Suicidal	82 ± 8	0.84 ± 0.09	0.81 ± 0.14	0.83 ± 0.07	0.91 ± 0.06

ROC-AUC Curves



- **Intention TOI stimuli** offer strongest discrimination of suicidal group from controls

Take-away:

- **DAR score:** Strongly identifies associations with depressive and suicidal ideation, with the "Intention" TOI alone effectively distinguishing suicidal risk, while combining TOIs enhances classification accuracy
- **DAR-score:** A promising tool for assessing mental health, capturing subtle mental associations through RTs for congruent and incongruent, offering insights that could support early intervention and treatment strategies.

References

- **Millner et al. (2018):** The brief death implicit association test: Scoring recommendations, reliability, validity, and comparisons with the death implicit association test. *Psychological Assessment*, pp. 1356-1366.
- **Barnes et al. (2017):** Moving beyond self-report: Implicit associations about death/life prospectively predict suicidal behavior among veterans. *Suicide and Life-Threatening Behavior*, pp. 67-77.

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